



Magnet RAM Capture Tool

◆ What It Does

- Captures **RAM contents** (running processes, network connections, passwords, encryption keys, malware in memory).
- Saves the dump file (.raw) for forensic analysis.
- Useful because RAM data is lost when the system is powered off.

◆ Steps to Use Magnet RAM Capture

1. Download & Prepare

- Get the tool from Magnet Forensics website.
- Save it on a **USB drive** (so you don't overwrite RAM on the target system).

2. Run on Target System

- Plug in USB and launch MagnetRAMCapture.exe.
- Select:
 - **Destination Folder** → where the memory dump will be saved.
 - **File Name** → give a unique name (e.g., incident123.raw).

3. Start Capture

- Click **Capture** → tool starts saving RAM data.
- Size depends on system RAM (e.g., 8GB RAM = ~8GB dump file).

4. Save Dump File

- Once finished, the .raw file will be in your chosen directory.
- Safely remove the USB drive.

5. Analyze the Dump

- Open .raw file in forensic analysis tools:
 - **Volatility Framework** (free) → process listing, network connections, DLLs, malware detection.
 - **Rekall** → another open-source memory forensics tool.
 - **Magnet AXIOM** → deeper integration with disk + memory evidence.



Analyzing RAM Dump (.raw) with Volatility

◆ Step 1: Install Volatility

- **Windows/Linux:** Download from [Volatility GitHub](#).
- **Volatility 2** (Python 2) or **Volatility 3** (Python 3).
- Works directly on .raw dump files.

◆ Step 2: Identify the Profile

Volatility needs to know the OS type. Run:

```
volatility -f memory.raw imageinfo
```

It will suggest the profile (e.g., Win10x64_18362).

◆ Step 3: Common Analysis Commands

1. List processes

```
volatility -f memory.raw --profile=Win10x64_18362 pslist
```

→ Shows running processes.

2. Check hidden/malicious processes

```
volatility -f memory.raw --profile=Win10x64_18362 psscan
```

3. Network connections

```
volatility -f memory.raw --profile=Win10x64_18362 netscan
```

→ Shows active connections (IP, ports).

4. Loaded DLLs (to spot injected code)

```
volatility -f memory.raw --profile=Win10x64_18362 dlllist -p <PID>
```

5. Dump suspicious process

```
volatility -f memory.raw --profile=Win10x64_18362 procdump -p <PID> -D ./output/
```

6. Search for passwords/keys

```
volatility -f memory.raw --profile=Win10x64_18362 mimikatz
```

(Sometimes works to extract creds.)



Alternative: Using Rekall

- Rekall is another memory forensics tool.
- Usage is similar:

```
rekall -f memory.raw pslist
```

```
rekall -f memory.raw netscan
```

◆ Typical Findings in RAM Forensics

- Malware injected in processes (e.g., svchost.exe, explorer.exe).
- C2 (Command & Control) IP addresses in netscan.
- Chat history, decrypted content, or passwords.



So the workflow is:

1. Capture RAM with Magnet RAM Capture (memory.raw).
2. Load file in **Volatility**.
3. Use commands to list processes, connections, DLLs, and dump suspicious files.